

PAPER_1356 — Active Matter Toner-Tu Universality

Framework: UQFF — Star-Magic v5.27+ | **Tier:** F Condensed Matter (CLOSED) **Calculator:**
calculate_paradox({"paradox": "active_matter"})

Master Expression

$\rho_{\text{flock}} = \beta_i \times \Phi_{\text{res}} = 0.506$; class via $D_{\text{phys}} - 1 = 3$

Match: STRUCTURAL

Interpretation

Flocking transition density driven by $\beta_i \times \Phi_{\text{res}}$. Universality class set by $D_{\text{phys}} - 1 = 3$ spatial dim.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$
- $\omega_{\text{SCm}} = 1.25$ THz

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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